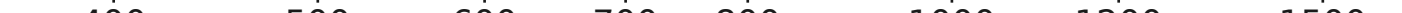
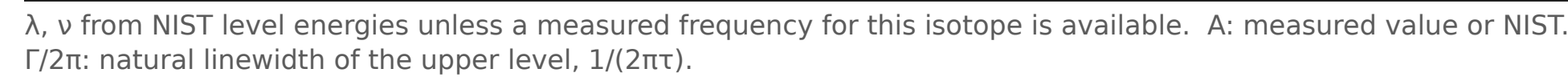


Zr⁺ ionisation limit 53507.83 cm⁻¹ = 6.63413 eV (186.889 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U er U \rangle $ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^3F_5 - y^3G_5^o$	468.9121	468.7809	639.3362	7.423	8.330e+07	13.26	100	proposed laser-cooling
$a^3F_5 - z^3G_5^o$	901.7653	901.5178	332.4506	≈ 1.665	$5.889e+05$	-	-	proposed narrow closed cooling line from metastable a^5F_5